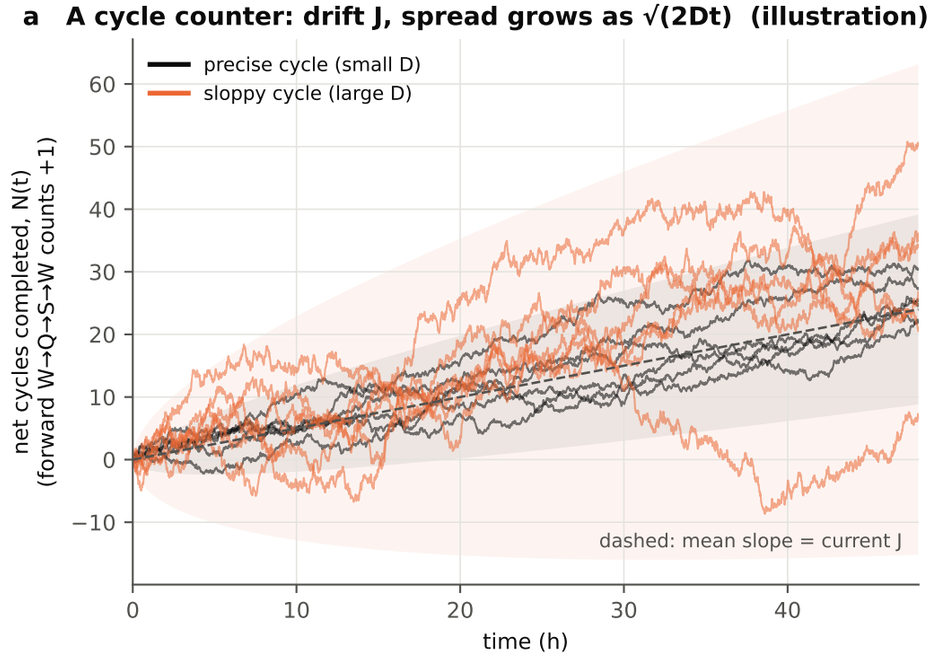
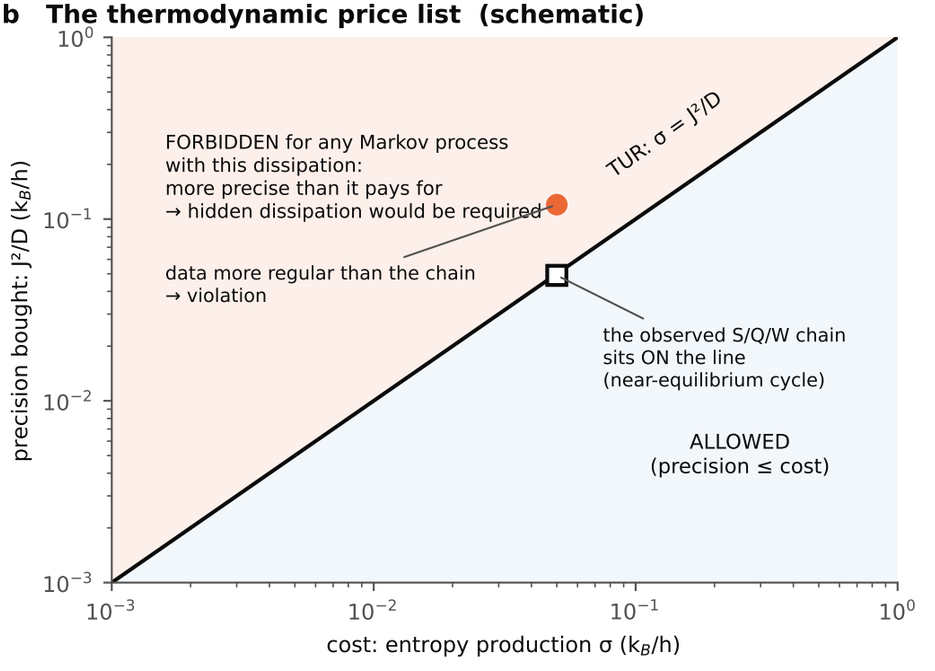
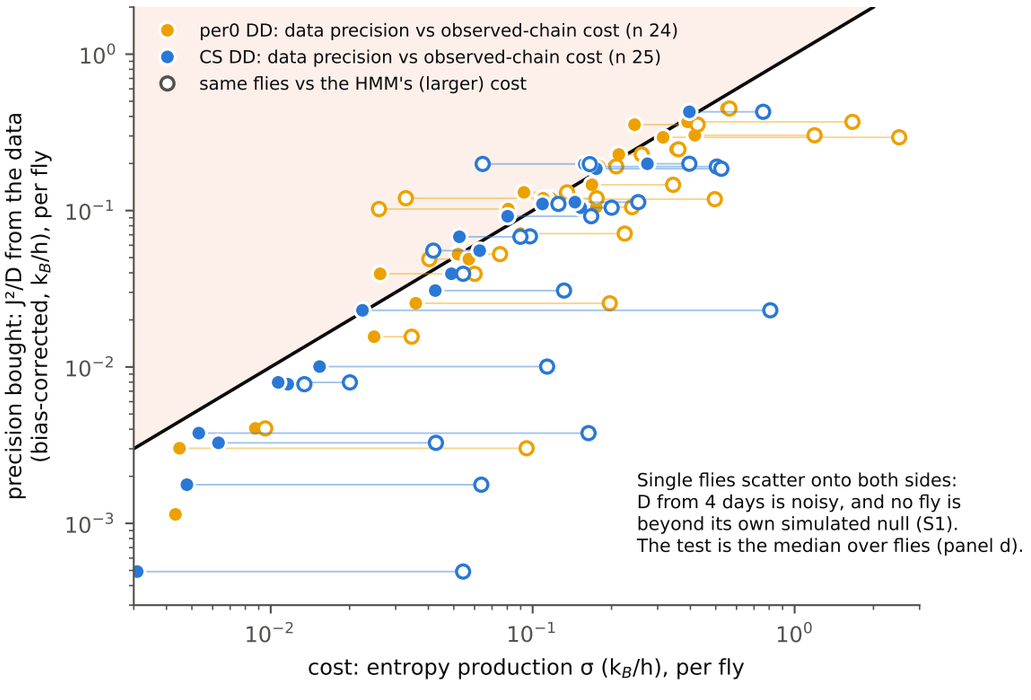
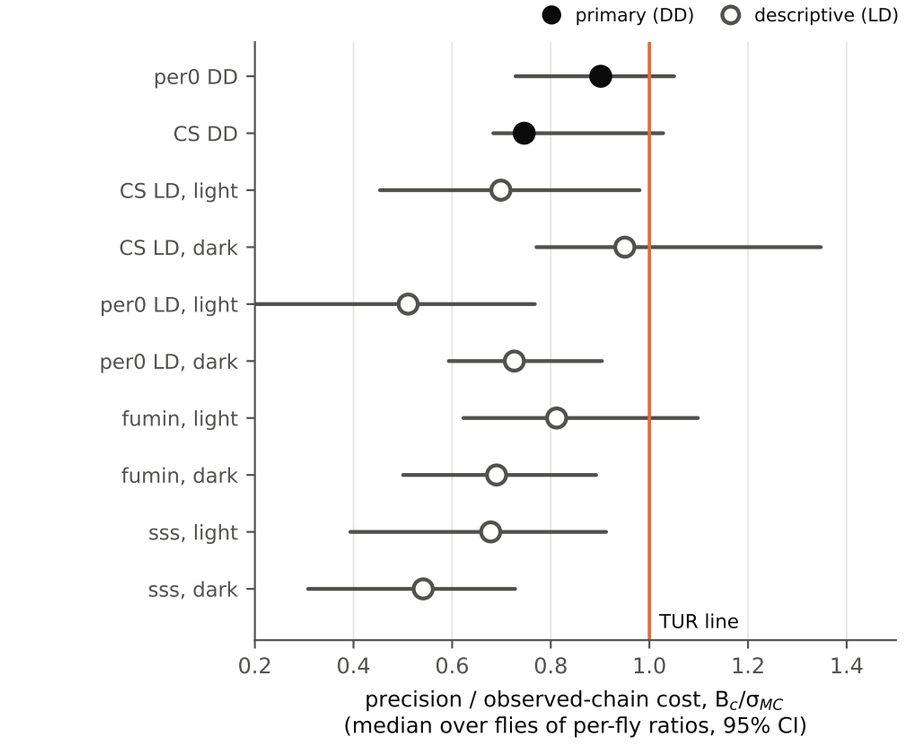








Reading: the cycle is never more regular than the observed chain's own dissipation allows (no hidden dissipation required). per0 DD sits close to the line;
CS DD and LD fall below it because slow modulation (free-running rhythm, day/phase drift) adds noise. Against the HMM's larger cost (open circles) the flies use only ~half.
Rule of three for one-way edges (fumin, sss) moves those LD ratios to 0.73-0.84, still below 1.